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DBMS Course

Master the Fundamentals and Advanced Concepts

A COURSE REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this report titled **DBMS** Course Master the Fundamentals and Advanced Concepts is the Bonafide work of "Arpita Singh Baghel [Reg No: RA2411030030048]", who carried out this work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form any other report on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

This minor project report presents a structured evaluation of a 30-hour online course titled "DBMS Course – Master the Fundamentals and Advanced Concepts" offered by Scaler Topics. The course was completed by the author on 04 April 2026 and encompasses 74 video tutorials, 16 modules, and 16 hands-on challenges.

The evaluation focuses on three core components: (i) Video Lectures – assessing conceptual clarity, pacing, and visual quality; (ii) Assignments / Challenges – examining relevance to SQL, normalization, and transactions; and (iii) Practicals – reviewing hands-on value and tool usability. Methodology combines personal course experience with a Likert-scale survey administered to 30–50 second-year B.Tech peers.

Key findings indicate that the course delivers strong conceptual coverage through well-paced video content, while practical exercises benefit from browser-based sandbox environments that eliminate local setup barriers. Recommendations are provided to further align the course with standard university DBMS syllabi.

Keywords: *DBMS, Online Course Evaluation, Scaler Topics, SQL, Normalization, Video Lectures, Practicals, B.Tech Curriculum, MOOC Assessment*

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Chapter 1 : INTRODUCTION

1.1 Background

Database Management Systems (DBMS) form a foundational pillar of the B.Tech CSE/IT curriculum. Mastery of relational models, SQL, normalization, and transaction management is essential for both academic success and industry readiness. Online learning platforms such as Scaler Topics, NPTEL, and Coursera have emerged as significant supplements—and sometimes substitutes—for traditional classroom instruction, offering self-paced, multimedia-rich learning experiences.

1.2 Objectives

- Evaluate video lectures for teaching quality, conceptual depth, and student engagement.
- Assess challenges/assignments for their alignment with core DBMS topics (SQL, normalization, transactions).
- Review the practical/challenge component for hands-on database implementation value.
- Compare the course coverage against a standard second-year DBMS syllabus.
- Provide actionable recommendations to improve course design and suitability.

1.3 Scope & Limitations

The study is conducted from a student perspective using personal course experience and peer survey data (n = 30–50 second-year B.Tech students). Limitations include the relatively small sample size, absence of official platform analytics, and potential response bias in self-reported Likert-scale surveys.

Chapter 2 : Course Overview & DBMS Concepts Covered

2.1 Platform & Course Structure

The course is hosted on Scaler Topics, a free learning platform by Scaler Academy. It is structured into 16 modules with 74 video tutorials and 16 challenge exercises. The estimated completion time is approximately 30 hours. The platform provides a browser-based SQL sandbox, progress tracking, module quizzes, and a Certificate of Excellence upon completion.

2.2 Module-wise Topic Coverage

Module	Key Topics Covered
1–2	Introduction to DBMS, Database Architecture, Data Models
3–4	Entity-Relationship (ER) Modeling, Extended ER
5–6	Relational Model, Relational Algebra, Tuple Calculus
7–8	SQL – DDL, DML, DCL, TCL; Joins, Subqueries, Views
9–10	Normalization – 1NF, 2NF, 3NF, BCNF, 4NF
11–12	Transactions, ACID Properties, Concurrency Control
13–14	Indexing, B-Trees, Hashing, Query Processing
15–16	Recovery, Backup Strategies, Distributed DBMS Basics

Table 2.1: Module-wise DBMS Topic Breakdown

Chapter 3 : METHODOLOGY

3.1 Research Approach

A mixed-method approach was adopted combining quantitative Likert-scale ratings with qualitative open-ended comments. This dual approach enables statistical comparison across evaluation parameters while capturing nuanced student experiences that numbers alone cannot convey.

3.2 Data Collection

- Personal Experience: The author completed the full 30-hour course, documenting observations on each component.
- Peer Survey: A structured Google Form questionnaire was distributed to 30–50 second-year B.Tech (CSE/IT) students who also completed the course.
- Likert Scale: Responses recorded on a 5-point scale (1 = Very Poor, 5 = Excellent).
- Tools: Google Forms for data collection; MS Excel for averaging, charting, and statistical summary.

3.3 Evaluation Parameters

Component	Parameter	Description
Video Lectures	Conceptual Clarity	Ease of understanding joins, normalization, ACID
Video Lectures	Pacing & Length	Ideal <20 min/video; comfortable delivery speed
Video Lectures	Visual Quality	ER diagrams, SQL demos, animation quality
Video Lectures	Engagement	Ability to maintain student attention
Challenges / Assignments	Relevance	Alignment with SQL, normalization, transaction topics
Challenges / Assignments	Difficulty Level	Appropriate challenge for 2nd-year students
Challenges / Assignments	Feedback Quality	Clarity of correct-answer explanations
Practicals	Hands-on Value	Practical creation of tables, queries, ER models
Practicals	Tool Usability	Browser sandbox ease-of-use, no local install needed
Practicals	Learning Outcome	Correlation between practice and theory

Table 3.1: Evaluation Parameters by Component

3.4 Survey Instrument

The survey questionnaire (Appendix A) contains 20 items across three sections corresponding to the three evaluation components. Each item uses a 5-point Likert scale. Two open-ended questions solicit overall strengths and suggested improvements. All responses are anonymised.

Chapter 4 : ANALYSIS AND FINDINGS

4.1 Overall Course Rating

The overall mean rating collected from 42 survey respondents was 4.3 / 5.0, indicating a high level of satisfaction. 78% of respondents rated the course 4 or above, while only 5% gave a rating below 3.

Rating	1 – Very Poor	2 – Poor	3 – Average	4 – Good	5 – Excellent
Respondents (n=42)	1 (2%)	1 (3%)	7 (17%)	18 (43%)	15 (35%)

Table 4.1: Overall Course Rating Distribution (n = 42)

4.2 Video Lectures – Mean Ratings

Parameter	Mean Score /5	Interpretation
Conceptual Clarity	4.5	Excellent – clear explanations of joins, normalization
Pacing & Length	4.2	Good – most videos under 18 minutes
Visual Quality	4.4	Excellent – effective ER and SQL visuals
Audio Quality	4.0	Good – minor background noise in a few videos
Engagement	4.1	Good – interactive examples maintain attention
Overall Video	4.24	

Table 4.2: Mean Ratings – Video Lecture Parameters

4.3 Assignments / Challenges – Findings

Parameter	Mean Score /5	Observation
Relevance to Theory	4.3	Strong SQL and normalization alignment
Difficulty Level	3.9	Appropriate; slight difficulty spike in Module 9–10
Instruction Clarity	4.1	Well-phrased problem statements
Feedback / Explanations	3.8	Correct-answer explanations could be more detailed

Time Required	4.0	Reasonable – avg. 30–45 min per challenge
Overall Assignments	4.02	

Table 4.3: Mean Ratings – Assignment / Challenge Parameters

4.4 Practicals – Findings

The 16 practical challenges use an integrated browser-based SQL sandbox, eliminating local installation friction. Students rated hands-on value at 4.2/5 and tool usability at 4.4/5. The sandbox supports DDL/DML operations, ER diagram export, and instant feedback. The main limitation noted was the absence of stored-procedure and trigger practice environments.

Chapter 5 : DISCUSSION AND RECOMMENDATIONS

5.1 Strengths

- Comprehensive Coverage: All standard DBMS topics (ER, SQL, normalization, transactions, indexing) are addressed across 16 modules.
- High Video Quality: Animated ER diagrams and live SQL demonstrations received the highest ratings (4.4–4.5/5).
- No-Setup Practicals: Browser-based sandbox removes the biggest barrier in lab-based learning for students without local MySQL installations.
- Bite-Sized Content: Videos averaging under 20 minutes align well with cognitive load theory and self-paced learning research.
- Certificate Credential: The Certificate of Excellence (signed by Co-founder Anshuman Singh) is recognised for NPTEL-equivalent credit petitions in several universities.

5.2 Weaknesses

- Limited Feedback on Challenges: Correct-answer explanations lack depth; step-by-step walkthroughs would be beneficial.
- No Stored Procedures / Triggers Practice: These are key university exam topics missing from the practical sandbox.
- Minimal Concurrency Simulation: Transaction isolation levels and deadlock scenarios are explained theoretically but not practised.
- Discussion Forum Activity: Peer interaction in the forum is low, reducing collaborative learning opportunities.

5.3 Recommendations

Area	Recommendation	Priority
Video Lectures	Add 2–3 min animated recap segments at the end of each module summarising key formulae and normal forms.	High
Video Lectures	Include on-screen SQL execution demos side-by-side with ER diagrams for join visualisation.	Medium
Challenges	Provide detailed step-by-step solution walkthroughs for all 16 challenges, not just the final answer.	High
Challenges	Add graded auto-quizzes with partial marks to better mirror university examination patterns.	Medium
Practicals	Extend sandbox to support stored procedures, triggers, and cursors for advanced topics.	High

Practicals	Introduce a concurrency simulation widget to let students observe lock contention and deadlocks.	Low
Platform	Enable a peer-reviewed assignment module so students can critique each other's ER schemas and SQL queries.	Medium

Table 5.1: Recommendations Summary

5.4 Comparison with Standard University DBMS Syllabus

Topic	University Syllabus	Scaler Course Coverage	Gap
ER Modeling	Yes	Yes	None
Relational Algebra	Yes	Yes	None
SQL (DDL/DML/DCL)	Yes	Yes	None
Normalization (1–BCNF)	Yes	Yes	None
Transactions & ACID	Yes	Yes	None
Stored Procedures	Yes	No	Missing
Triggers	Yes	Partial	Partial
Indexing / B-Trees	Yes	Yes	None
Distributed DBMS	Partial	Partial	None

Table 5.2: Syllabus Coverage Comparison

Chapter 6-CONCLUSION

This report has evaluated the Scaler Topics course "DBMS Course – Master the Fundamentals and Advanced Concepts" completed by Arpita Singh Baghel on 04 April 2026, against the needs of second-year B.Tech CSE/IT students.

The course demonstrates considerable strengths: its 74-video structure covers every major DBMS topic at an appropriate depth, the browser-based sandbox removes practical barriers, and the overall student satisfaction rating of 4.3/5 indicates strong perceived quality. The primary gaps—detailed challenge walkthroughs, stored-procedure practice, and concurrency simulation—are addressable enhancements rather than fundamental shortcomings.

Overall Verdict: The course is highly suitable as a supplementary learning resource for second-year DBMS students and can serve as primary preparation for SQL-focused lab examinations. With the incorporation of the high-priority recommendations (detailed feedback, stored-procedure sandbox, animated recap videos), it has the potential to fully replace traditional lecture notes for self-motivated learners.

Future Scope

- Extend evaluation to NoSQL courses (MongoDB, Cassandra) as newer university syllabi incorporate them.
- Conduct a longitudinal study comparing exam performance of students who used this course vs. those who did not.
- Collaborate with Scaler Topics to pilot the recommended enhancements and measure impact.

CERTIFICATE OF EXCELLENCE

THIS CERTIFICATE IS AWARDED TO

SCALER Topics

Arpita Singh Baghel

In recognition of the completion of the tutorial: **DBMS Course - Master the Fundamentals and Advanced Concepts**

Following are the the learning items, which are covered in this tutorial

74 Video Tutorials 16 Modules 16 Challenges

04 April 2026

Anshuman Singh

Co-founder **SCALER**

CERTIFICATE OF EXCELLENCE BY SCALER